Automated Recognition and Classification of Trading Cards

Presentation of the Unassisted project

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Motivation and Objective

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The problem

- Manual sorting of trading cards (TCG) is time-consuming and prone to errors.
- Existing commercial solutions (e.g., CardBot) are proprietary and/or expensive.
- Many different similar cards. MTG has >100k unique cards.

Objectives of the work

- Software: Recognition pipeline that recognizes cards even from poor-quality photos.
- Hardware: Affordable and widely available components (Arduino, stepper motors...).
- The entire project FOSS and well documented so that it can serve as educational material.

Hardware: Concept

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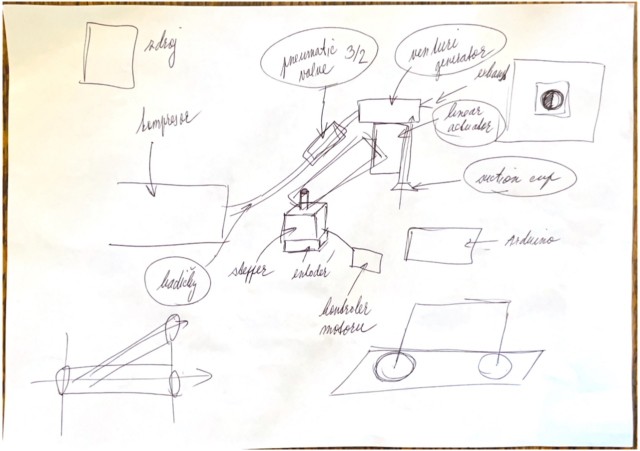
- Focus on affordable components (FOSS philosophy).
- Control logic implemented on Arduino Platform.
- **Milestone reached:** Precise stepper motor control using rotation encoders feedback loop.



Data Acquisition & Augmentation

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Synthetic Dataset Generation

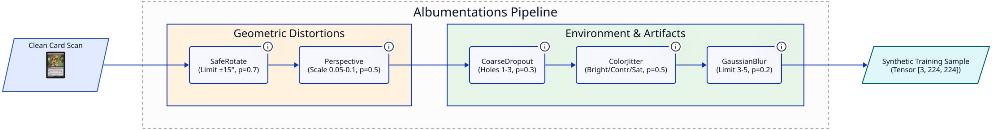
- Real-world labeled dataset is non-existent.
- Scans retrieved via ***Scryfall API***.
- Custom naming convention for human-readability & uniqueness:

Bridge the Domain Gap

- Heavy augmentation pipeline required to match camera output.
- **Geometric:** Random perspective, rotation.
- **Photometric:** Blur, noise, lighting conditions, partial occlusions.

```
filename = edition_collectorNumber_name-UUID.png
zen_21_kor-outfitter-00006596-1166-4a79-8443-ca9f82e6db4e.png
```

Augmentation Pipeline



Softmax vs ArcFace

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Limitations of Standard Softmax

- Huge number of classes $(10^5+) \rightarrow$ computationally expensive.
- Open-set problem: New cards are released frequently.

Metric Learning Approach (ArcFace)

- Learning a mapping to a hypersphere embedding space.
- Intra-class ***compactness***. Images of the same card are pulled together.
- Intra-class ***separability***. Different cards are pushed apart.
- Inference via ***Cosine Similarity*** (Nearest Neighbor).

Geometric Interpretation

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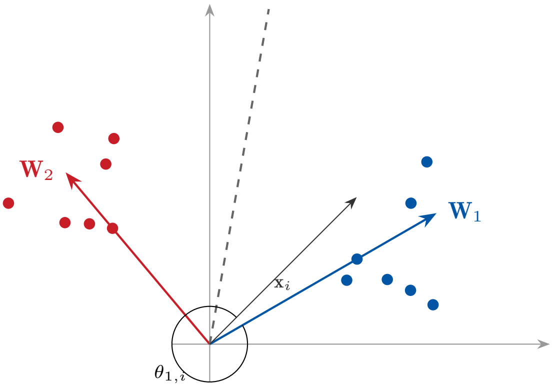
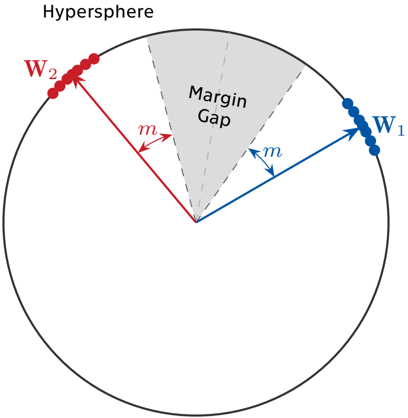


(a) Standard Softmax Loss



(b) ArcFace Loss

Recognition Pipeline

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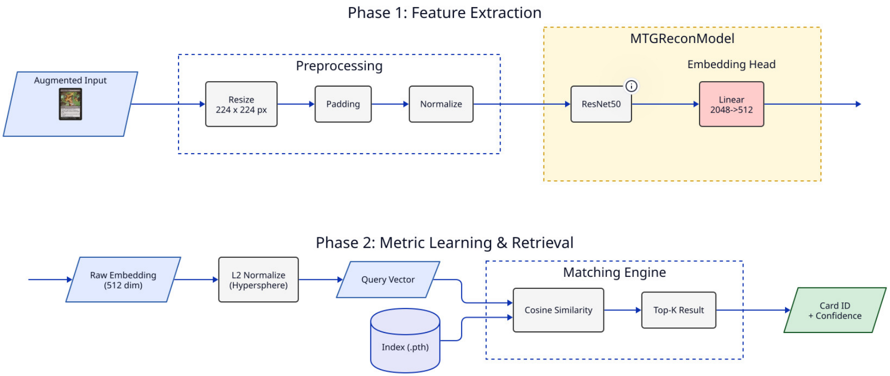

Conclusion and Future Work

Summary

- Robust recognition pipeline implemented (Synthetic Data + ArcFace).
- Basic HW control operational.

Next Steps

- Finalize mechanical construction (feeding mechanism).
- Full integration of the recognition module with the robotic control.
- Final evaluation on a physical test set.